

Bookly Customer Support Agent

Grounded, clarify-first AI support for an online bookstore · github.com/andrewwchild/booklydemo · booklydemo.streamlit.app

Thesis

The best CX agents don't guess — they clarify, then act on real data. Human in conversation, machine-precise on facts. Transactional queries always use tools; the LLM never invents order IDs, tracking numbers, or stock levels.

Architecture

Layer	Role
Chat UI	Streamlit (demo) · FastAPI + CLI alternates
Orchestrator	Thin LLM ↔ tool loop · rules fallback · no LangChain
Memory	Per-session slots: order_id, reason, email, book_title
Tools (7)	lookup_order · verify_identity · initiate_refund · check_stock · get_policy · send_password_reset · escalate_to_human
Data	JSON stores (prototype) → OMS / inventory / CRM in production
Quality	30-case golden-set eval harness · 100% pass · CI-ready

Key Design Decisions

- **Tools over RAG** for transactions — zero hallucination on order/stock/refund data
- **Clarify-first** — ask for order ID, email, reason before acting
- **Verify before refund** — identity check, then eligibility check, then initiate
- **Human handoff** — escalation ticket + full conversation summary
- **Dual engine** — LLM when API key set; deterministic rules for demo + evals

Live Demo Highlights

Flow	Input	Proves
Order lookup	"Where's my order?" → ORD-1001	Clarify-first + grounded tracking
Refund	ORD-1003 · damaged · carol@example.com	Multi-turn slots + identity verify
Edge case	ORD-1004 · david@example.com	Verification ≠ approval
Escalation	"Speak to a real person"	Ticket + transcript handoff
Inventory	Fourth Wing / Atomic Habits	Live stock via tool, not generation

Production Roadmap

Gap Today	Production Fix
JSON data stores	OMS + inventory + CRM APIs
Email-match identity	Auth session + OTP
Keyword escalation	Confidence threshold + sentiment
In-process sessions	Redis + cross-device continuity